

1st SIT Coursework - 02 Question paper**Year Long 2025/2026**

Module Code:	CC5002NI
Module Title:	Software Engineering
Module Leader:	Mr. Rubin Thapa (Islington College)

Coursework Type:	Individual
Coursework Weight:	This coursework accounts for 35% of the overall module grades.
Submission Date:	30 th April, 2026
Coursework given out:	Week 19
Submission Instructions:	Submit the following to the Islington College's MST portal before 06:00 PM on the due date: • A report (document) in .pdf format in the MST portal or through any medium which the module leader specifies.
Warning:	London Metropolitan University and Islington College takes plagiarism very seriously. Offenders will be dealt with sternly.

PLAGIARISM

You are reminded that there exist regulations concerning plagiarism. Extracts from these regulations are printed overleaf. Please sign below to say that you have read and understand these extracts:

Extracts from University Regulations on Cheating, Plagiarism and Collusion

Section 2.3: *“The following broad types of offence can be identified and are provided as indicative examples*

- (i) *Cheating: including taking unauthorised material into an examination; consulting unauthorised material outside the examination hall during the examination; obtaining an unseen examination paper in advance of the examination; copying from another examinee; using an unauthorised calculator during the examination or storing unauthorised material in the memory of a programmable calculator which is taken into the examination; copying coursework.*
- (ii) *Falsifying data in experimental results.*
- (iii) *Personation, where a substitute takes an examination or test on behalf of the candidate. Both candidate and substitute may be guilty of an offence under these Regulations.*
- (iv) *Bribery or attempted bribery of a person thought to have some influence on the candidate's assessment.*
- (v) *Collusion to present joint work as the work solely of one individual.*
- (vi) *Plagiarism, where the work or ideas of another are presented as the candidate's own.*
- (vii) *Other conduct calculated to secure an advantage on assessment.*
- (viii) *Assisting in any of the above.*

Some notes on what this means for students:

1. Copying another student's work is an offence, whether from a copy on paper or from a computer file, and in whatever form the intellectual property being copied takes, including text, mathematical notation, and computer programs.
2. Taking extracts from published sources *without attribution* is an offence. To quote ideas, sometimes using extracts, is generally to be encouraged. Quoting ideas is achieved by stating an author's argument and attributing it, perhaps by quoting, immediately in the text, his or her name and year of publication, e.g. “ $e = mc^2$ (Einstein 1905)”. A *reference* section at the end of your work should then list all such references in alphabetical order of authors' surnames. (There are variations on this referencing system which your tutors may prefer you to use.) If you wish to quote a paragraph or so from published work then indent the quotation on both left and right margins, using an italic font where practicable, and introduce the quotation with an attribution.

CONTRACT CHEATING

Contract cheating (also known as assessment outsourcing, commissioning or ghost writing) is when someone seeks out another party, or AI generator service, to produce work or buy an essay or assignment, either already written or specifically written for them or the assignment to submit as their own piece of work.

Contract cheating undermines the integrity of the academic process and devalues the qualifications awarded by the university. Students are reminded that academic integrity is a fundamental principle of our institution. Engaging in contract cheating not only impacts the individual's academic record but also the reputation of the university.

Students are encouraged to seek support if they are struggling with their coursework. The university offers a range of resources, including academic counseling, tutoring services, and workshops on study skills and time management. Utilizing these resources can help students achieve their academic goals without resorting to dishonest practices.

Penalty:

- Failure in the Module: The student must re-register for the same module, and the re-registered module will be capped at a bare pass.
- Ineligibility to Continue on the Course: Where re-registration of the same module, or a suitable alternative, is not permissible, the student will not be able to continue on the course. Additionally, the following penalty will be applied to the student's final award:
 - Undergraduate Honors: The student's final classification will be reduced by one level.
 - Unclassified Bachelors: Downgraded to Diploma in Higher Education.
 - Foundation Degree: Distinction downgraded to Merit; Merit downgraded to Pass; Pass downgraded to Certificate in Higher Education.
 - Masters: Distinction downgraded to Merit; Merit downgraded to Pass; Pass downgraded to Postgraduate Diploma.

Reporting and Consequences:

Instances of contract cheating will be thoroughly investigated, and students found guilty will face the penalties outlined above. It is the responsibility of every student to ensure that their work is their own and to avoid situations that could lead to accusations of academic misconduct.

By adhering to these standards, students contribute to a fair and equitable academic environment, ensuring the value and recognition of their qualifications are maintained.

CS5002NI Software Engineering

Assignment - (Individual)

1. Introduction

This assignment contributes 35% to the overall mark for this module.

2. Client Background

Gokyo Bistro is a popular restaurant located at Jhamsikhel, Lalitpur. They have a very refreshing ambience with the provision of both outdoor and indoor dining. Recently they have been facing problems in their regular operations of table arrangement and order. There has been a massive surge in customers nowadays. A lot of customers have to return due to maximum occupancy. Also, a lot of customers have increased asking for home delivery. So, they are looking for an online system that can manage all these operations. Assuming you and your team have been approached for this project. You are required to carry out Object Oriented Analysis and Design for this project for below given specifications.

DETAILED SPECIFICATION

The proposed system is to have following functions

1. *Login and Registration*

The system should have 3 types of users. Admin, visitors and members. Admin and Members can login with their respective credentials.

2. *Book Table*

Visitors/Members can check the availability of tables and reserve the table for a particular date/time.

3. *Set Beverage choices*

Though not compulsory, while booking, visitors can set the expected beverages choices they might prefer from the available list whereas premium members can reserve their wine/whiskey choices.

4. *Payment*

There is a requirement to pay 1200 advance in order to reserve the table. Reservation can be cancelled before the final day whereas in case the reservation is cancelled on the very day, a fee of 500 is deducted from advance before returning the amount to the visitors/members.

5. *Generate Report*

Admin should have a facility to generate the financial report. An advanced mechanism is needed for admin to predict the revenue for further years along with the recommendation of menu items.

6. *Bill Generation*

There should be the privilege to generate bills for admin and generate receipts.

7. *Order placement*

Members can place orders for food and beverages with an option of home delivery. Once again payment can be made online as well.

8. *Update Menu*

Admins can add menu items, delete and update menu items.

9. *Rate the experience*

Visitors/members can rate the overall experience of food, staff's behavior , ambience and feedback can also be provided.

10. *Post offers and schemes*

Admin should be able to post various special offers and discount schemes from time to time.

Questions

1. Prepare a Work Breakdown Structure as a part of Scope Identification. Also prepare a Gantt Chart showcasing the schedule of work with a reasonably assumed timeline. But please ensure that the Gantt chart is based upon a methodology of your choice and the activities/tasks should be relevant to the methodology chosen.
2. Produce a Use Case Model for the required system – including a Use Case diagram, and High-Level Use Case Descriptions for each Use Case. For two of the Use Cases of your choice, produce Expanded Use Case Descriptions.

Choose one of these expanded Use Case Descriptions to model Collaboration or Sequence diagrams, in 3, below.

3. Produce a Collaboration diagram or a Sequence diagram for the chosen use case description as mentioned above.
4. Draw an activity diagram for the use case description that hasn't been chosen for the Sequence/Collaboration diagram above.
5. Prepare a table comprising all possible domain classes from all the respective use cases. Produce an Analysis Class diagram showing the domain classes. Your Diagram should include at least 2 inheritance, 2 aggregation and 2 composition relation and should have necessary association and multiplicities showcased.
6. Several tasks such as analysis and design diagrams have already been completed. How do you plan to proceed further in order to complete the project? Briefly explain your plan in words. While explaining the plan, you are required to select a development methodology that you think is feasible for this project, and your explanation of the subsequent steps should be related to the methodology you have chosen.

(Note: the further planning should compulsorily include following things:

- 1) Architectural choice.
- 2) Design Pattern.
- 3) Development Plan (that includes tools/resources/programming platforms that are going to be used and priority order of artifacts/features that you have planned tentatively.
- 4) Testing Plan.
- 5) Maintenance Plan.

7. Develop some design prototypes for the proposed system for features which are listed in detailed specification.

(Note: the prototypes should cover the major features and at least 15 separate prototypes (UI) need to be shown.

Marking Scheme

Individual Marks	Marks
WBS and Gantt Chart	20
Use Case	10
High Level Use Case	5
Expanded Use Case	5
Collaboration Diagram or Sequence Diagram	10
Activity Diagram	10
Class Diagram	15
Further Development	10
Prototype	15

MILESTONE BREAKDOWN

MILESTONE 1 (April 06, 2026 6:00 pm)

- All tasks except Further development portion (q.no 6) in a single pdf file.

FINAL SUBMISSION (April 30, 2026 6:00pm)

- All tasks in a single pdf report with answer number clearly written in the report body along with Introduction and Conclusion.

END